An Evidence-Aware Statistical-Mechanics Atlas for Modern Machine Learning

Taxonomy, finite-size diagnostics, causal interventions, and cross-domain stress tests

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Abstract

Statistical-mechanics language is increasingly used to describe modern machine learning, but "phase," "entropy," and "size" often refer to incompatible objects and ensembles. We introduce an evidence-aware atlas that separates seven scientific coordinates—system, state object, deformation, disorder ensemble, scale path, observable, and phenomenon—from the protocol used to test them and from the evidence allowed to support a claim. A machine-readable PhaseCard binds each question to an estimator, null, intervention, holdout, finite-size diagnostic, and falsifier.

The empirical program contains a five-seed Transformer screen (8,700 trained models), a frozen twelve-seed confirmation suite (1,632 models), a public 45-run GPU reference grid, and controlled cross-domain stress tests. The GPU grid shows a finite-width learning response: at its largest registered point, semantic order is 0.188 ± 0.004 and normalized IID risk is 0.812 ± 0.004. However, it has no interior susceptibility peak, no Binder crossing, no attention specialization, and one collapsed control pair; it therefore does *not* support a phase-boundary claim. Frozen confirmations instead support narrower statements about feed-forward interventions, optimizer-conditioned risk, data-scaling paths, and natural-data measurement. Diffusion, reinforcement-learning, and multi-agent cases validate the grammar at model level, not a shared universality class. The result is a falsifiable map from scientific questions to bounded claims, including negative and unresolved outcomes rather than a collection of visually suggestive curves.

Scope, reading path, and headline result

This paper is about using numerical statistical mechanics to audit collective behavior *inside learning systems*. It is not a survey of machine learning for discovering phases of physical matter, and it does not claim that every learning curve has a thermodynamic limit. Readers interested in the conceptual contribution can follow Sections 2–4; readers interested in estimators and finite-size logic can continue through Sections 5–6; and readers interested in the measurements can start at Sections 7–8.

The headline empirical result is deliberately negative. Increasing width and data improves a small Transformer, and matched MLP ablation has a measurable effect, but the reference grid fails the registered tests needed for phase language. The finding is *learning response observed; phase boundary not established*. This bounded conclusion is the intended behavior of the atlas, not a failure to produce a positive plot.

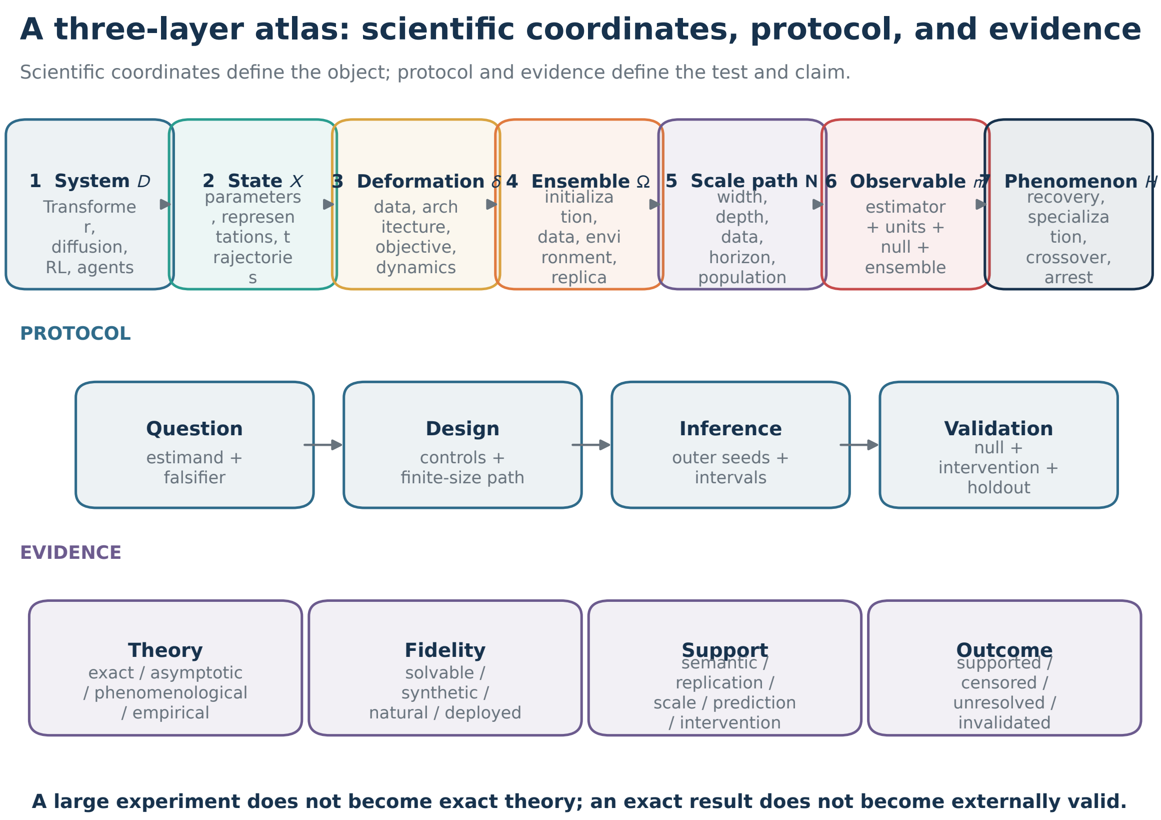

Figure 1: The atlas has three layers. Seven scientific coordinates specify what object is under study. A protocol specifies how the object is measured, perturbed, and held out. Independent theory, fidelity, support, and outcome fields specify what may be claimed. Collapsing these layers is the main source of ambiguous phase language in machine-learning experiments.

1 Introduction

Statistical mechanics supplies machine learning with a precise language for ensembles, collective observables, response, scale, and limiting behavior. It has clarified solvable inference thresholds, information–computation gaps, signal propagation, jamming, and optimization landscapes [1, 15, 33]. Modern architectures add a harder numerical question. A Transformer has attention and feed-forward modules, several scale coordinates, non-equilibrium training dynamics, and data-dependent semantics; diffusion and reinforcement learning replace a static parameter ensemble with trajectories and feedback. A curve can be steep in all these systems without representing the same physical object or the same kind of transition.

Current numerical work is fragmented along three boundaries. First, *ontology* is often implicit: "size" may mean width, depth, data, context, population, or trajectory resolution. Second, *measurement* is often underspecified: temporal checkpoint variance can be mislabeled as disorder susceptibility, and token entropy can be mistaken for strategy or macrostate entropy. Third, *evidence* is often compressed into a single adjective: an exact result in a restrictive model, a five-seed synthetic screen, and a natural-data measurement may all be called a "phase transition" despite answering different questions.

We ask one central question:

> How can numerical statistical mechanics organize and falsify collective phenomena across modern learning systems while keeping the state object, ensemble, finite-size path, and evidential scope explicit?

Our answer is an evidence-aware atlas. Its core is a seven-coordinate scientific taxonomy. Around that core, a protocol layer records estimators, disorder, controls, interventions, and holdouts; an epistemic layer records theory status, fidelity, evidence support, and outcome. This separation makes a negative result first class: a boundary may be censored, a response may remain finite-width, an order parameter may lose semantic validity, or two conditions may be incommensurate.

The empirical focus is intentionally asymmetric. Transformers receive broad module, optimizer, scaling, and data screens followed by frozen confirmations. A smaller GPU reference grid exposes raw outer-seed distributions and tests the phase-claim decision rule. Diffusion, reinforcement learning, and multi-agent systems are cross-domain stress tests for the grammar. They are not asserted to share an equilibrium ensemble or critical exponent. A separate controlled predictive-continuation simulator checks software and holdout semantics; because it generates its own response law, it is method validation rather than evidence about trained neural systems.

The paper makes five contributions:

1. a precise seven-coordinate taxonomy and a thirteen-outcome vocabulary, with protocol and evidence axes kept outside the scientific ontology;
2. an observable dictionary that binds every estimator to units, ensemble, null, validity range, and allowed interpretation;
3. a finite-size claim rule requiring semantic validity, independent disorder, scale diagnostics, robustness, and a prospective check;
4. a unified empirical report covering 8,700 exploratory and 1,632 frozen-confirmatory Transformer runs, plus an auditable 45-run GPU reference grid; and
5. an explicit claim ledger that distinguishes observed learning response, bounded mechanism evidence, model validation, negative phase evidence, and open external-validity cells.

The main scientific conclusion is narrower than the atlas coverage. The reference Transformer learns more with width and effective data, and MLP ablation matters increasingly at larger width. Yet no registered finite-size diagnostic supports a phase boundary on that grid. This is the kind of distinction a statistical-mechanics study must preserve.

2 Related work and positioning

We organize prior work by the question it answers, not by a flat list of model families. This also separates the present problem—statistical mechanics *of* learning systems—from machine learning used as a tool for classifying physical phases.

2.1 Solvable learning and inference

The statistical mechanics of learning begins from a specified ensemble and a limit in which quenched disorder and macroscopic order parameters become tractable [10, 23]. Replica, cavity, state-evolution, and related methods identify information-theoretic thresholds, algorithmic thresholds, and computationally hard regions in planted inference problems [33]. These works establish the standard we adopt: a transition is attached to a state object, a disorder law, a scale, and an observable. Our atlas does not replace those theories. It records which of their assumptions survive when a trainable implementation removes a solvable constraint.

2.2 Deep networks: signal propagation, kernels, and landscapes

Statistical-mechanics analyses of deep learning cover expressivity, signal propagation, random matrices, optimization landscapes, jamming, and nonequilibrium dynamics [1]. Mean-field signal propagation separates ordered and chaotic regimes at initialization [30]; neural tangent kernels describe a wide-network lazy limit [18]; exact deep-linear dynamics show how representations and timescales emerge during training [29]; and the jamming perspective relates interpolation and loss geometry [15]. Random-feature asymptotics clarify double descent and generalization in another controlled limit [22].

These contexts are complementary rather than interchangeable. Initialization signal propagation is not a terminal trained-model ensemble; kernel regression is not feature learning; interpolation is not semantic recovery; and a finite-width optimizer trajectory is not automatically an equilibrium sample. The atlas therefore records theory status and state object separately.

2.3 Modern architectures and driven systems

For attention, a solvable tied low-rank model exhibits a positional–semantic transition [6], and attention-indexed models extend Bayes-optimal analysis to full-width matrices [5]. The unresolved transport question is whether order semantics and a boundary persist under standard feed-forward blocks, normalization, causal objectives, optimization, and natural carriers. Width, depth, context, and data are distinct deformations; compute-optimal scaling and hyperparameter transfer do not make them one thermodynamic coordinate [17, 32].

For diffusion, the published oracle-score analysis distinguishes early speciation from later collapse toward memorized samples [4]; learned DDPM and DiT endpoints add score error, architecture, and solver effects [16, 26]. For reinforcement learning, reward-model overoptimization demonstrates why proxy reward, gold reward, KL pressure, and strategy support must be measured jointly [13, 28]. For multi-agent systems, classical consensus dynamics and modern MARL or language-agent interactions add field, coupling, topology, and population coordinates [8, 9, 21]. Recent 2026 claims about diffusion and collective language-agent behavior are treated as hypotheses rather than established anchors [2, 7, 34].

2.4 Reproducibility and the gap addressed here

Model cards, datasheets, and reproducibility checklists make assumptions and artifacts inspectable [14, 24, 27]. Preregistration separates exploratory choices from confirmatory tests [25]. None of these alone supplies a statistical-mechanics ontology: they do not specify which randomness defines susceptibility, which component is the finite-size coordinate, or when a threshold is censored. Conversely, a phase diagram alone does not encode holdouts, provenance, or invalidation. The present work joins these two traditions through a typed PhaseCard and an evidence vector.

The positioning is therefore precise. We contribute neither a new universal critical exponent nor a comprehensive survey of every physics-inspired machine learning theory. We contribute an operational taxonomy for numerical studies of modern learning systems and demonstrate it on a deliberately heterogeneous set of artifacts.

3 Taxonomy: what is being studied?

The taxonomy classifies a *scientific object*, not a software module and not a level of evidence. A registered atlas point is the seven-tuple

$$\mathcal{A} = (D, X, \delta, \Omega, \mathbf{N}, m, H), \tag{1}$$

where D is the system, X the state object, δ the deformation, Ω the disorder ensemble, $\mathbf{N}$ the scale vector and path, m the observable, and H the hypothesized phenomenon. Two studies can be compared only after the components that are asserted equivalent have been named.

3.1 The seven scientific coordinates

1. System D. The system fixes the update law and interaction structure: Transformer, diffusion process, reinforcement learner, or multi-agent population in this release. A domain label is a root node, not a claim that all descendants share one ensemble.

2. State object X. Examples include trained parameters, hidden representations, attention maps, feed-forward activations, optimizer trajectories, generated samples, policy occupancy, reward trajectories, beliefs, messages, and agent actions. A single run may expose several state objects, but each observable names the one on which it is defined.

3. Deformation δ. A deformation changes data or environment, architecture or interaction, objective or feedback, dynamics or optimizer, or lifecycle. Width is a scale coordinate, whereas replacing a feed-forward activation is an architectural deformation; conflating them makes a scaling statement ambiguous.

4. Disorder ensemble Ω. Initialization, data draw, minibatch order, environment instance, evaluation sample, dropout, diffusion noise, rollout, and inner replica are distinct random variables. The independent outer unit used for uncertainty must be declared.

5. Scale path $\mathbf{N}$. Scale is a vector such as $(d, L, n, T, C, A, \ldots)$ for width, depth, data, context, compute, and population. A finite-size study declares one coordinate as N and registers how all others move or remain fixed. "Larger model" is not a scaling path.

6. Observable m. An observable is an estimator together with units, normalization, ensemble, null, and validity range. Normalized risk, semantic order, susceptibility, Binder cumulant, participation ratio, attention entropy, causal risk difference, gold reward, and truth-conditioned consensus are different observables even when all are dimensionless.

7. Phenomenon H. The hypothesis may concern recovery, specialization, locality, multiplicity, symmetry breaking, crossover, criticality, glassiness, dynamical arrest, speciation, reward collapse, or collective alignment. A phenomenon is not assigned from visual slope; it is adjudicated by registered diagnostics.

3.2 What is deliberately outside the taxonomy

Theory status (exact, asymptotic, phenomenological, empirical), fidelity (solvable, trainable synthetic, natural, deployed), and evidence support (semantic, replicated, finite-size, predictive, interventional, external) are orthogonal metadata. This prevents an exact solvable result from being called externally valid, and prevents a large natural-data experiment from being called exact theory. Protocol fields—question, estimator, null, calibration, holdout, intervention, and falsifier—are also separate because the same atlas point can be tested well or poorly.

3.3 Outcome taxonomy

An atlas edge receives one of thirteen non-ordinal outcomes. Transport outcomes are preserved, renormalized, rounded, or censored. Topology outcomes are split, merged, or new regime. Accessibility outcomes are hysteretic, path-dependent, or statistical–computational separation. Evidential failures are semantic failure, unresolved, or not comparable. These labels answer *what happened* before stronger phase language is considered.

Coordinate	Transformer	Diffusion	Reinforcement learning	Multi-agent
State X	representations, attention, MLP, optimizer path	score/sample trajectory, endpoint	occupancy, policy, reward path	beliefs, messages, actions, population
Deformation δ	data, block, objective, optimizer, lifecycle	score, schedule, solver, guidance	environment, verifier, KL, update rule	field, coupling, topology, memory, roles
Disorder Ω	initialization, data, minibatch, evaluation	data, score noise, reverse noise	environment, rollout, verifier noise	private evidence, topology, update order
Scale $\mathbf{N}$	width, depth, data, context, steps	resolution, NFE, data, model size	horizon, samples, verifier calls	population, messages, rounds
Observable m	semantic order, risks, spectra, interventions	speciation, nonlocality, commitment, proximity	gold/proxy reward, support, entropy	truth consensus, polarization, accuracy
Phenomenon H	recovery, specialization, crossover	speciation, collapse, nonlocality	productive specialization, reward collapse	alignment, fragmentation, error amplification

Table 1: Domain crosswalk for six coordinates. The system coordinate is the column header. Entries are registered possibilities, not a claim that every cell has been run.

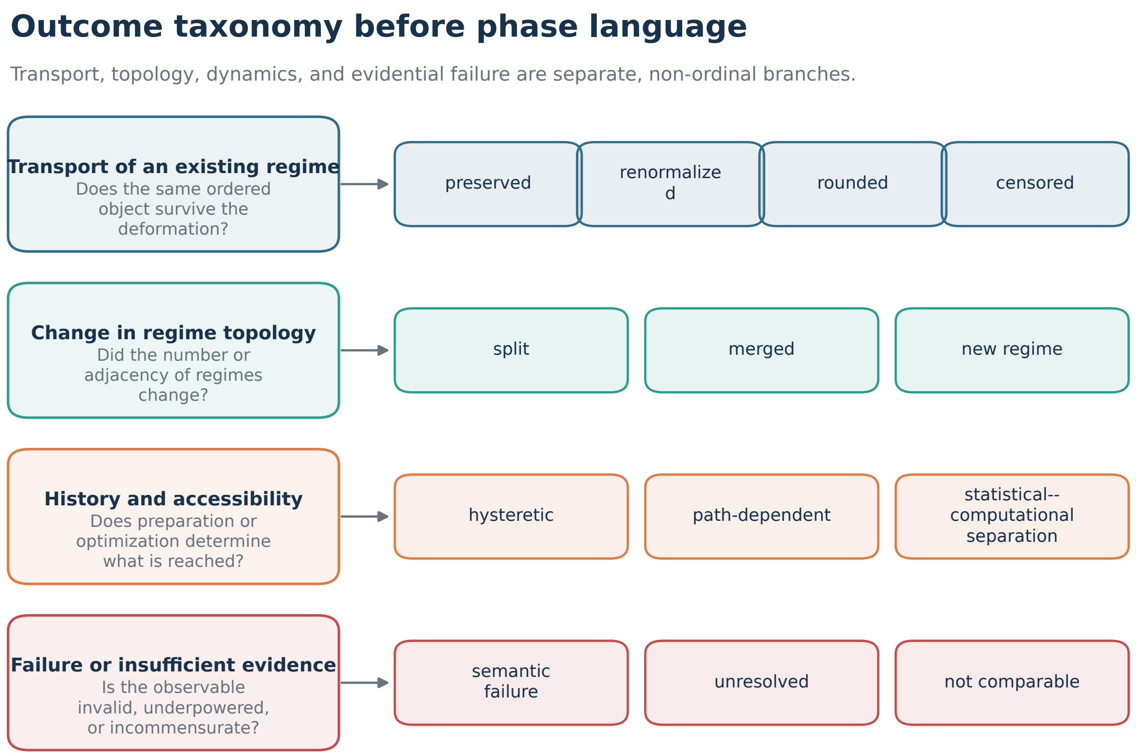

Figure 2: The registered outcome vocabulary. Transport, topology, accessibility, and evidential failure are separate branches. "Unresolved" records insufficient separation; "not comparable" records a broken common estimand; neither is silently dropped from the atlas.

RQ	Question	Required diagnostic	Main artifact	Status
1	Does an order parameter retain semantic meaning?	latent/shuffled probe; separate IID and OOD risk	observable dictionary and reference grid	**supported**
2	Does a response sharpen into a finite-size boundary?	interior peak or crossing; window and size sensitivity	GPU reference diagnostics	**negative**
3	Which module or interaction causally supports task order?	matched ablation; nonvanishing normalization; fresh seeds	MLP confirmation	**bounded**
4	How do optimizer and scale path change the learned state?	frozen selection slice; compute and cap audit	optimizer and scaling confirmations	**bounded**
5	Can a calibrated relation predict an untouched condition?	nontrivial baselines; disjoint holdout; censoring	controlled continuation simulator	model validation only
6	Which conclusions survive a natural or learned endpoint?	comparable estimand, disjoint split, independent implementation	corpus and cross-domain dossiers	partial; mostly open

Table 2: Question-to-claim map. "Bounded" means that the registered contrast is supported while a neighboring stronger interpretation remains unestablished.

4 Research questions and claim map

The paper is organized around six questions. Each question has a primary estimand, a diagnostic that could block the claim, and an evidence status. This structure prevents the experiments from appearing as an unrelated plot catalog.

The reporting template is the same for every result: *observation* states the estimate; *interpretation* states the narrow reading justified by its null and evidence vector; *not established* blocks a tempting stronger claim; and *next falsifier* identifies the experiment that could change the status. Sections 8– 9 follow this order.

Observable dictionary: estimator + ensemble + null

Family	Representative estimator	Averaging ensemble	Required control	Allowed reading
Risk	$R_{train}, R_{IID}, R_{OOD}$	evaluation examples	chance / calibrated baseline	prediction quality
Recovery	$I(m; Z)/H(Z)$	outer seeds + examples	shuffled latent labels	semantic order
Specialization	head / role assignment	permutation-aligned seeds	permuted components	symmetry breaking
Response	$\partial\langle m\rangle/\partial g$	independent outer seeds	window + scale sensitivity	crossover candidate
Fluctuation	$N\,\mathrm{Var}_{\Omega}(m)$	outer disorder only	temporal variance	susceptibili ty
Dynamics	time-to-order, path entropy	training trajectories	matched compute	accessibilit y / arrest
Mechanism	$R_{ablated} - R_{full}$	matched interventions	equal-resource sham	causal contribution

Rule: without ensemble, units, validity range, and null, an estimator is not a registered observable.

Figure 3: Observable dictionary used throughout the atlas. Risk, semantic recovery, specialization, response, fluctuation, dynamics, and intervention effects require different averaging ensembles and nulls. Similar numerical ranges do not make the estimands interchangeable.

5 Method: from atlas point to falsifiable experiment

5.1 PhaseCard: one typed experimental record

The machine-readable PhaseCard joins the scientific tuple in Eq. (1) to a protocol and an epistemic record:

$$\mathcal{P} = (\underbrace{D, X, \delta, \Omega, \mathbf{N}, m, H}_{\text{scientific object}}, \underbrace{g, \mathcal{I}, \mathcal{N}, \mathcal{H}, \mathcal{F}}_{\text{protocol}}, \underbrace{T, L, \mathbf{E}, C}_{\text{epistemic record}}). \tag{2}$$

Here g denotes controls, $\mathcal{I}$ interventions, $\mathcal{N}$ nulls, $\mathcal{H}$ holdouts, and $\mathcal{F}$ falsifiers. T is theory status, L fidelity, $\mathbf{E}$ a vector of evidence coordinates, and C the outcome. Resources, software identity, seed streams, and artifact hashes are attached as execution metadata; changing them does not silently change the scientific identity.

This representation enforces several invariants. Inner stochastic replicas are not counted as independent outer disorder. A threshold has an explicit crossing or censor status. A metric definition includes units and normalization. A post-hoc selected coordinate cannot be labeled preregistered. A failed accelerator attempt and a fallback are distinct execution attempts under one scientific condition.

5.2 An observable is more than a scalar

Every reported estimate has the form

$$\widehat{m} = (\bar{m}, [m_-, m_+], q, U, n_{\text{outer}}, n_{\text{inner}}, \Omega, \text{units}, \text{status}), \tag{3}$$

where q is interval level and U names the uncertainty procedure. The status records validity and censoring. This prevents a temporal standard deviation, a between-initialization variance, and a bootstrap interval from sharing the same unqualified field name.

5.3 Primary observables

Intensive predictive risks. Cross entropy is normalized by the random-reference entropy, $R_{\text{CE}} = \ell_{\text{CE}}/\log V$, and Brier risk by $1-1/V$ for vocabulary size V. Train, IID, and OOD risks are stored separately:

$$R_{\text{train}}, \qquad R_{\text{IID}}, \qquad R_{\text{OOD}}, \qquad e_{\text{gen}} = R_{\text{IID}} - R_{\text{train}}. \tag{4}$$

Tokens, FLOPs, wall time, parameter count, and examples are resources rather than extensive versions of the risk.

Semantic order. When a latent variable Z is known, semantic retention is measured through a registered probe, conceptually

$$S_{\text{sem}} = I(m; Z)/H(Z), \tag{5}$$

and checked against shuffled labels. In the public reference grid the stored semantic order is the signed task-success coordinate produced by the controlled retrieval task. It remains a task order, not a universal magnetization.

Outer-disorder response and Binder diagnostic. For independent outer-seed terminal orders m_s and declared finite-size coordinate N,

$$\chi_N = N\left(\langle m^2\rangle_\Omega - \langle m\rangle_\Omega^2\right), \qquad U_4 = 1 - \frac{\langle m^4\rangle_\Omega}{3\langle m^2\rangle_\Omega^2}. \tag{6}$$

Checkpoint fluctuations from one training trajectory are named temporal variance and never substituted into Eq. (6).

Causal module effect. For matched full and ablated evaluations, the raw effect is $\Delta R = R_{\text{ablated}} - R_{\text{full}}$. A bounded signed effect uses a non-vanishing registered risk scale. Ratios that divide by remaining accuracy near zero are invalid because they can diverge without a physical change.

5.4 Outcome assignment and finite-size logic

A phase-language claim must pass five gates: semantic validity; an independent disorder ensemble; an interior finite-size signature such as a crossing, sharpening response, or stable collapse; robustness to fitting window and minimum size; and a frozen prospective check. Failing a gate produces a named negative outcome rather than an omitted figure.

5.5 Predictive phase continuation

Predictive continuation is one protocol inside the atlas. Given calibration observables $\mathcal{O}_{\text{cal}}$ and a surrogate $\mathcal{O}_{\text{solv}}(\theta)$, fit

$$\widehat{\theta}_{\text{eff}} = arg\min_{\theta} D\big(\mathcal{O}_{\text{cal}}^{\text{target}}, \mathcal{O}_{\text{solv}}(\theta)\big), \tag{7}$$

and evaluate only on withheld quantities,

$$\mathcal{E}_{\text{bridge}} = D\Big(\mathcal{O}_{\text{holdout}}^{\text{target}}, \mathcal{O}_{\text{solv}}(\widehat{\theta}_{\text{eff}})\Big). \tag{8}$$

Constant, source-boundary, size-only, nearest-calibration, and base-surrogate predictors are mandatory alternatives. Crossed, left-censored, and right-censored thresholds are retained separately. The controlled simulator in this release tests this accounting logic; it does not establish transport in a real training stack.

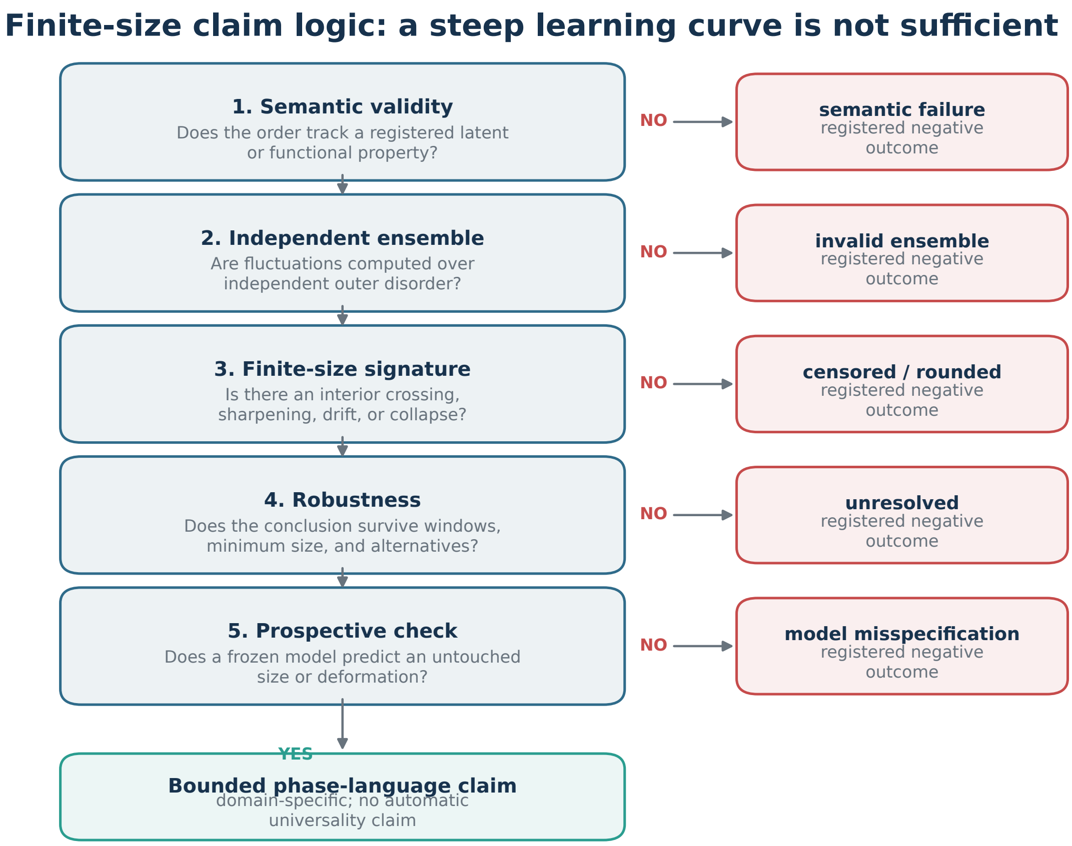

Figure 4: Decision rule for finite-size claims. Each failed gate has a specific negative outcome. Passing the sequence licenses only a domain-specific, bounded claim; it does not by itself establish universality or an equilibrium thermodynamic limit.

Design	Independent unit	Replication	Selection rule	Allowed language
Broad tensor screen	trained model	5 seeds / condition	exploratory	candidate effect, finite-grid response, negative result
Frozen confirmation	trained model	12 fresh seeds / condition	fixed from screen audit	registered contrast, bounded mechanism evidence
GPU reference grid	trained model	5 seeds / condition	portable pilot	outer-seed estimate; no asymptotic phase claim
Predictive simulator	outer disorder profile	16+16 seeds, 4 inner	split before fitting	software/method validation
Natural endpoint	trained byte model	12 fresh seeds	disjoint byte ranges	measurement transport, corpus-specific result

Table 3: Statistical roles of the experiment families. Replication count is one coordinate of evidence and never overrides an invalid order parameter or a missing finite-size signature.

6 Statistical contract

6.1 Nested disorder and uncertainty

A condition is a unique PhaseCard. A trained model under one outer seed is one independent unit. Inner stochastic evaluations are conditional replicates and are averaged within that unit before uncertainty is computed. For S outer seeds with means $\bar{m}_s$,

$$\bar{m} = \frac{1}{S}\sum_{s=1}^{S}\bar{m}_s, \qquad \widehat{\mathrm{SE}}(\bar{m}) = \sqrt{\frac{1}{S(S-1)}\sum_{s}(\bar{m}_s - \bar{m})^2}. \tag{9}$$

Five- and twelve-seed Transformer summaries use two-sided 95% Student-t intervals over complete full-pipeline seeds. The predictive simulator uses outer-profile resampling while preserving paired control curves. No inner replica is promoted to an outer sample.

6.2 Exploration, confirmation, and multiplicity

The five-seed tensor is an exploratory screen. It locates candidate contrasts, invalid controls, and negative outcomes; it does not support a universal phase claim and no familywise hypothesis-test interpretation is attached to its many intervals. The confirmation suite uses fresh seeds and freezes the selected module variants, optimizer slices, data paths, and corpus splits before those seeds are aggregated. Confirmation upgrades only the listed contrast, not every neighboring claim.

6.3 Finite-size fitting and censoring

Finite-size scaling is domain specific [3, 11, 12]. We do not impose $N^{-1/2}$ or N^{-1} on width, trajectory resolution, horizon, and population as universal exponents. Candidate drift and response-width models must be selected below a registered largest size, which is then used for prospective scoring. A curve that never crosses a threshold is left- or right-censored, not assigned the nearest endpoint as a critical point.

Sensitivity analysis varies the minimum included size, fitting window, threshold definition, and plausible alternative functional forms. A phase claim additionally requires order semantics and a finite-size diagnostic that is not generated by a duplicated control, resource cap, or evaluation leak.

6.4 Completeness and invalidation

Strict aggregation requires exactly the manifest seed set, finite required metrics, compatible trajectory checkpoints, and identical metric schemas. It never intersects whichever keys happened to survive. Each artifact records raw outer-seed values, nested seed identities, requested and effective resources, cap-hit and censor flags, configuration and data hashes, and elapsed time. Changing an estimand, split, or selection rule invalidates the old artifact for that claim without deleting its provenance.

7 Experimental design

7.1 Study A: broad Transformer tensor

The exploratory tensor contains 8,700 trained small decoder models across 1,740 conditions and five full-pipeline seeds. It varies feed-forward type, residual and normalization choices, objective interpolation, optimizer geometry, width/depth/context/data scaling paths, and synthetic-to-natural data carriers. The screen records train, IID, and OOD risk; semantic task order; attention and MLP interventions; gradient and update geometry; activation occupancy and spectral diagnostics; trajectory statistics; and resource accounting. Its purpose is breadth and failure discovery.

7.2 Study B: frozen Transformer confirmations

The confirmation suite contains 1,632 models in 136 conditions with 12 fresh seeds per condition. Four feed-forward variants are compared with matched parameter budgets where applicable. Optimizer learning rates are selected on calibration slices and frozen for evaluation. Data-scaling paths remove the cap artifact found in the screen. Natural-corpus byte ranges are non-overlapping and report bits per byte together with train–test and cross-corpus diagnostics.

7.3 Study C: portable GPU reference grid

The public reference study is intentionally small enough to audit at raw-seed level: widths $d \in \{16, 32, 64\}$, requested sample coefficients $\alpha \in \{0.5, 2, 8\}$, and 5 independent full-pipeline seeds, for 45 trained models. Each one-layer pre-normalized Transformer uses four heads, a GELU feed-forward block with ratio 2, AdamW, 24 update steps, and a controlled retrieval task. It records 81 aggregate fields, including separate risks, causal ablations, gradient geometry, attention entropy, susceptibility, Binder cumulant, and raw outer-seed samples.

An audit found that the input pipeline enforces at least 32 training examples. Consequently, $(d, \alpha) = (16, 0.5)$ and $(16, 2)$ both use 32 examples and are the same effective control point. They remain visible as a duplicated point and are not counted as two independent locations on a response curve. This finding motivates the dense confirmatory design in Appendix D.

7.4 Study D: cross-domain and continuation stress tests

Controlled diffusion, reinforcement-learning, and multi-agent simulators vary reverse time/locality, KL pressure/verifier noise, and field/coupling, respectively. Their role is to test whether the PhaseCard can express trajectory, feedback, and population observables without reusing Transformer semantics. The predictive-continuation benchmark contains many simulator tasks and nested replicas, but because the response model is generated by the same controlled family, it validates holdout and censoring mechanics only.

7.5 Compute and portability

All trainable runs are specified independently of site paths. Repository, data, manifest, interpreter, and output locations are runtime inputs. GPU use is an execution coordinate, not part of the scientific identity. Public artifacts do not contain hostnames, scheduler partitions, user directories, or private data locations.

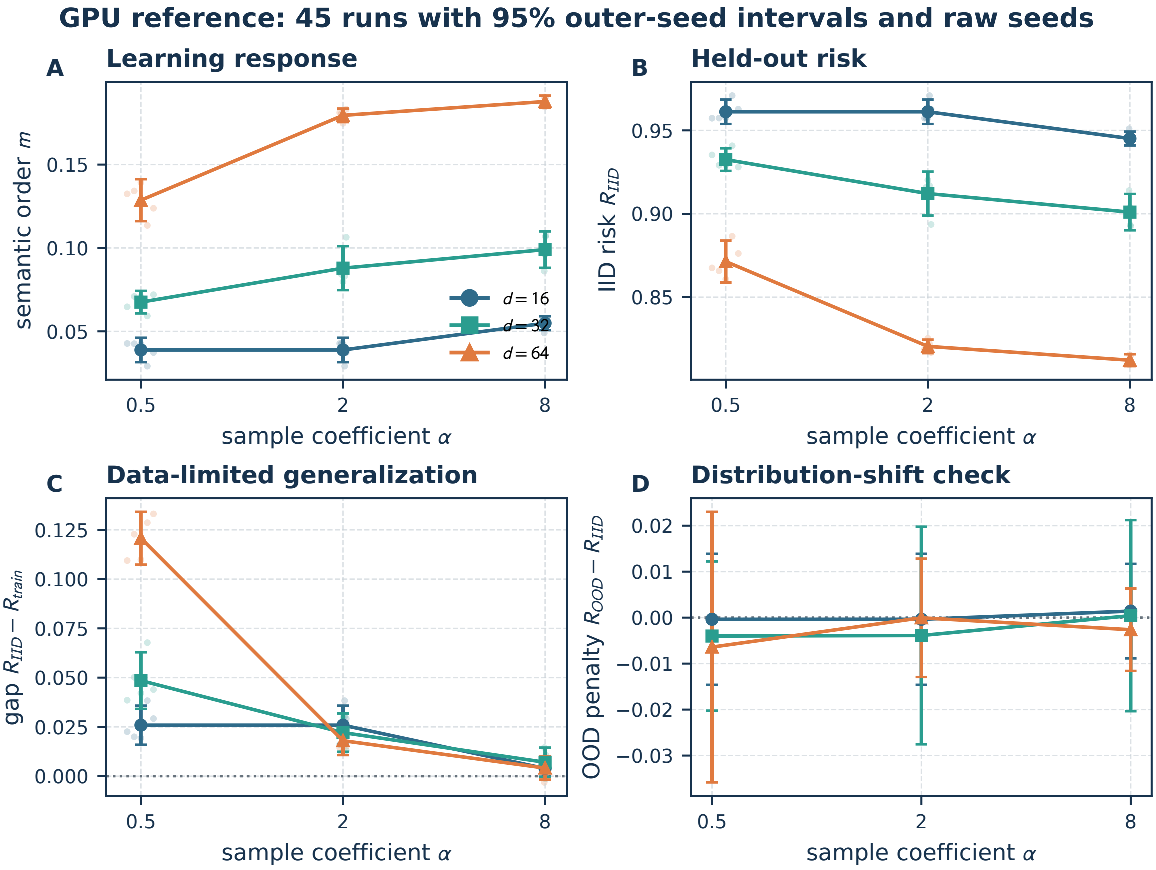

Figure 5: Public GPU reference study. Points and lines are condition means with two-sided 95% Student-t intervals over five independent full-pipeline seeds; translucent points are the raw seed values. (A) Semantic task order. (B) Normalized IID risk. (C) IID–train generalization gap. (D) OOD–IID difference with a conservative interval formed from the two marginal intervals. The nominal control is shown, while Figure 7 exposes the duplicated effective point.

8 Results: Transformer measurements and falsifiers

8.1 RQ1: width and effective data produce a learning response

Observation. Figure 5A shows raw outer-seed task order and 95% intervals. At $d = 16$, the duplicated 32-example controls produce identical aggregates, after which order increases modestly at 128 examples. Across width, the response is stronger: at $(d, \alpha) = (64, 8)$, semantic order is 0.188 ± 0.004 and IID risk is 0.812 ± 0.004. The resource coordinate, not the nominal control label, explains the duplicated point.

Interpretation. The registered small Transformer exhibits finite-width learning and benefits from effective data. This validates the task order as a responsive measurement on the tested grid.

Not established. A monotone improvement is not a transition. The grid has only three widths, one duplicated point, and 24 optimization steps. It does not identify a thermodynamic scale path or asymptotic threshold.

Next falsifier. Use distinct effective sample counts, longer training, five sizes with the largest held out, and a dense control grid fixed before the new seeds are observed.

8.2 Generalization and OOD behavior are separate observables

At width 64, the low-data generalization gap is 0.121 ± 0.013, whereas the high-data gap is 0.004 ± 0.006. The OOD-minus-IID differences in Figure 5D are small relative to conservative summed intervals across this particular synthetic shift.

The result is a data-limited overfitting signature, not an OOD robustness claim. The OOD generator shares the registered task family, and the five-seed intervals are too wide to rank small penalties. Reporting train, IID, and OOD risks separately prevents a reduction in one from being described as a general improvement in all three.

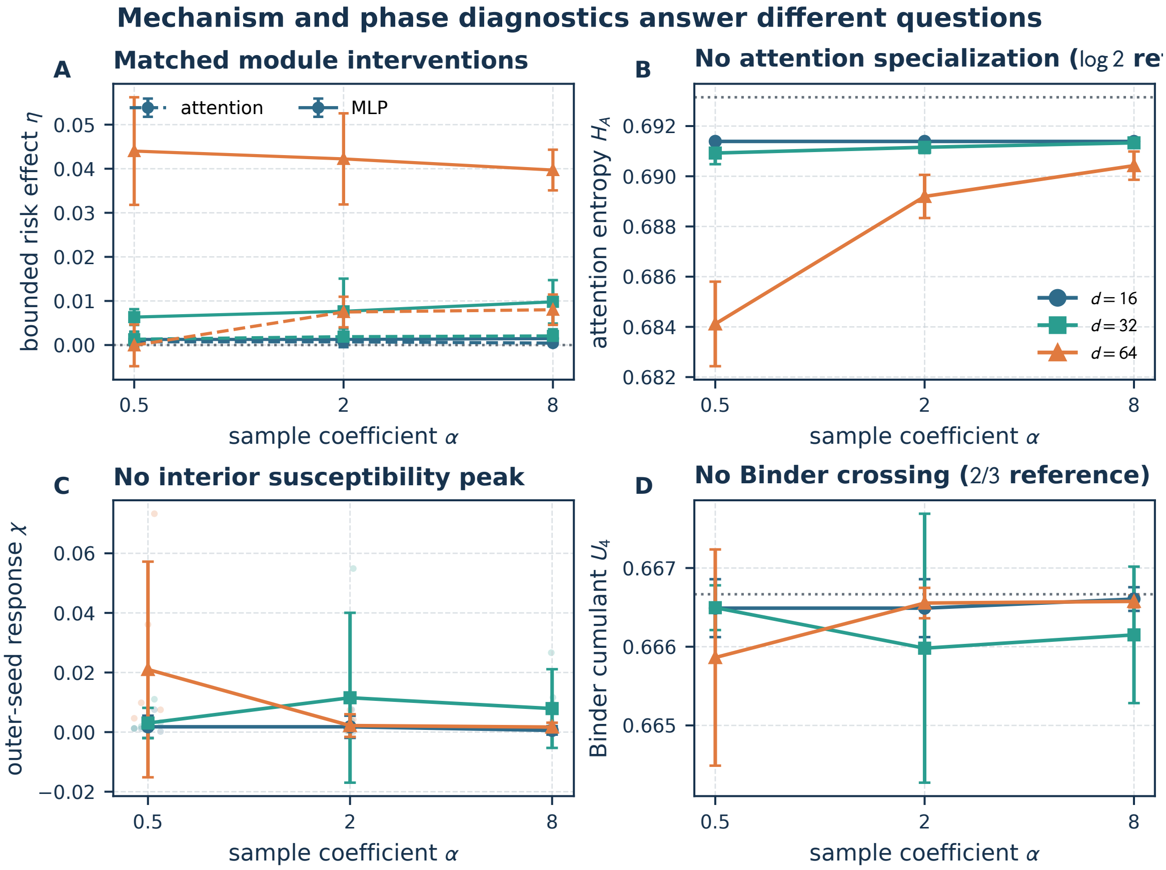

Figure 6: Mechanism and phase diagnostics from the same reference models. (A) Bounded matched-ablation effects for attention (dashed) and MLP (solid). (B) Attention entropy with log 2 reference. (C) Outer-seed susceptibility; the largest-width peak is an endpoint. (D) Binder cumulant with 2/3 reference. The causal effect in panel A does not upgrade the negative finite-size evidence in panels B–D.

8.3 RQ2: registered phase diagnostics reject a boundary claim

Observation. The largest-width susceptibility is maximal at the left endpoint rather than at an interior control. Response peaks do not exhibit a stable drift sequence. Binder cumulants remain near $2/3$ without a crossing, and attention entropy remains close to $\log 2$ rather than showing head specialization (Figure 6B–D).

Interpretation. The reference grid supports the outcome *finite-width learning response; phase boundary not supported*. Its evidence vector has semantic and outer-seed coordinates, but lacks finite-size and prospective-prediction support.

Not established. Neither criticality, universality, a positional– semantic transition, nor a Binder crossing is observed. The absence of evidence on this small grid is not evidence that larger or longer-trained Transformers cannot transition.

Next falsifier. The dense confirmation should test whether an interior peak emerges after removing effective-control duplication and optimization censoring. If it does not, the response should remain classified as rounded or censored.

8.4 RQ3: module interventions reveal a width-dependent MLP effect

Observation. In the reference grid the largest registered MLP effect is 0.044 ± 0.012, while the largest attention effect at the high-data, largest-width point is 0.008 ± 0.003. MLP effects grow with width, while attention effects remain smaller on this shallow, short-trained retrieval task (Figure 6A).

Interpretation. The feed-forward path causally contributes to the registered endpoint under matched ablation. The effect is an $O(1)$ risk difference with outer-seed uncertainty.

Not established. This does not show that MLPs dominate attention in general, that the modules define separate phases, or that the effect is stable under depth, long training, or natural language.

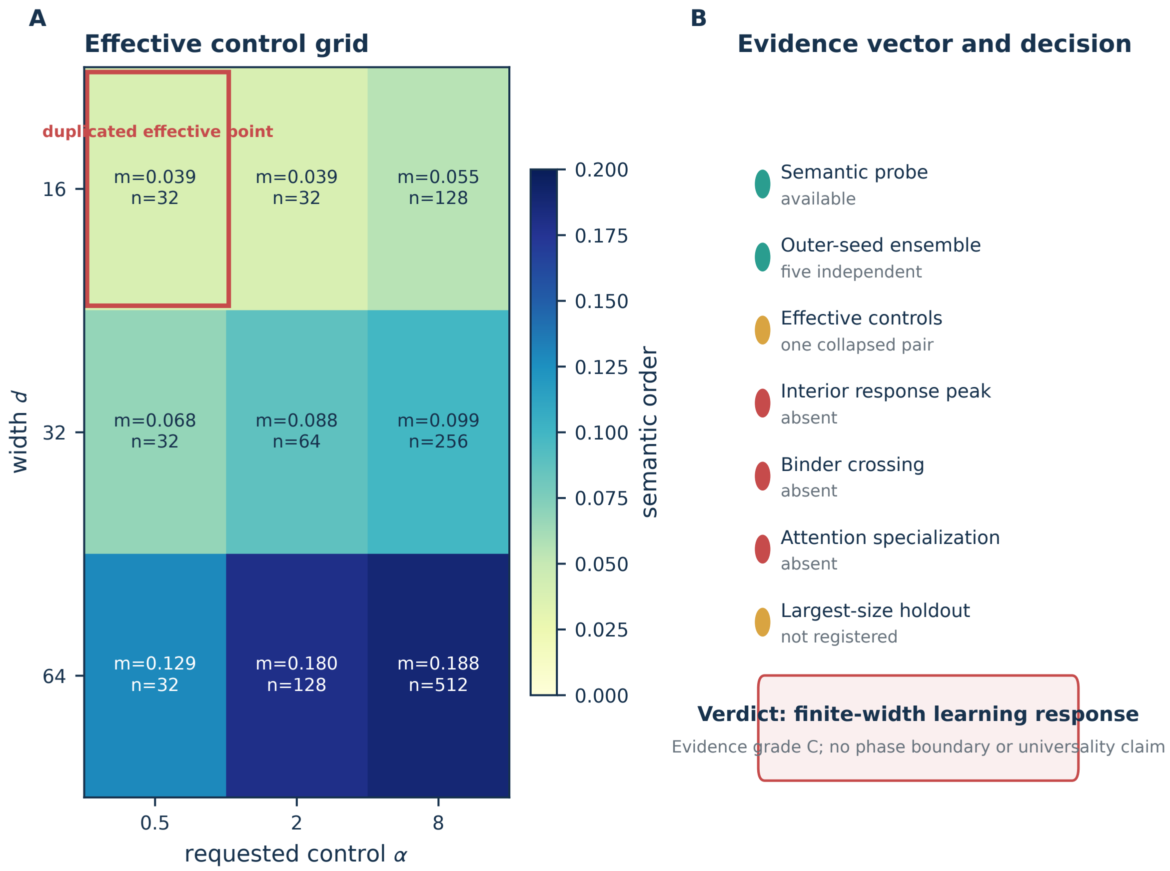

Figure 7: Audit and evidence verdict. (A) Semantic order and effective training-example count for every reference condition; the red outline identifies two nominal controls that collapse to the same 32-example point. (B) Evidence-vector components. The result is retained as a grade-C finite-width learning response, not converted to a phase boundary.

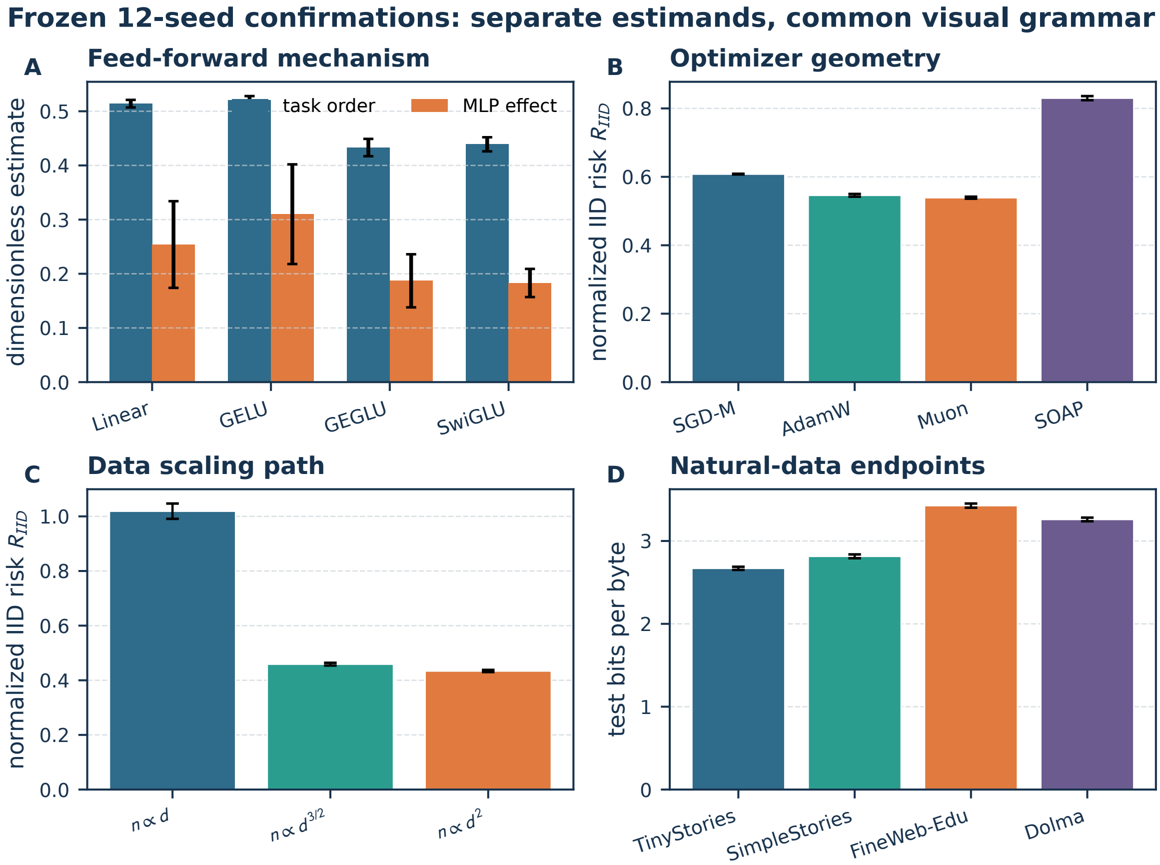

Figure 8: Unified summary of the frozen twelve-seed confirmations. Error bars are 95% Student-t intervals over fresh full-pipeline seeds. (A) Task order and matched MLP causal effect. (B) Optimizer-conditioned IID risk on frozen slices. (C) Uncapped data-scaling paths. (D) Disjoint-range natural-corpus bits per byte. Panel values are separate estimands and are not pooled into a single score.

8.5 RQ3–RQ4: frozen confirmations bound stronger conclusions

The twelve-seed suite tests hypotheses selected from the broad screen. Linear and GELU blocks reach task orders 0.514 ± 0.007 and 0.521 ± 0.007, compared with 0.433 ± 0.016 and 0.439 ± 0.013 for matched gated variants. Their bounded MLP effects are 0.254 ± 0.080, 0.310 ± 0.092, 0.187 ± 0.049, and 0.183 ± 0.026. The task therefore separates these registered architectures, but the causal intervals do not justify a universal ranking of gating mechanisms.

On frozen learning-rate slices, normalized IID risks are 0.608 ± 0.001 (SGD with momentum), 0.546 ± 0.004 (AdamW), 0.539 ± 0.003 (Muon), and 0.830 ± 0.006 (SOAP). These are optimizer-conditioned outcomes inside the registered grid. They are not a general optimizer benchmark because update scale, compute, and hyperparameter budgets are not exhaustively matched [19, 20, 31].

The uncapped data paths produce normalized risks 1.019 ± 0.028, 0.459 ± 0.005, and 0.434 ± 0.004 for $n \propto d$, $d^{3/2}$, and d^2 at the largest common endpoint. Natural-data byte models produce 2.670 ± 0.019, 2.815 ± 0.023, 3.427 ± 0.025, and 3.260 ± 0.022. Differences across corpora mix modeling difficulty and corpus statistics; they validate a common bits-per-byte measurement, not a corpus quality ranking.

9 Cross-domain dossiers and coverage

Cross-domain coverage tests whether the taxonomy can preserve different state objects and observables. It does not make the domains coequal in evidential depth (Figure 9).

9.1 Diffusion trajectories

The state object is a reverse-time sample trajectory. Reverse time is the primary control; locality or score deformation is secondary. Registered observables include speciation, nonlocality, trajectory commitment, and proximity to memorized samples. The oracle-score anchor separates speciation from later collapse [4]; learned DDPM and DiT models would add score and architecture error [16, 26].

Observation. The controlled dossier can represent separate speciation and collapse coordinates and retain censored boundaries. **Interpretation.** This validates the trajectory PhaseCard and observable namespaces. **Not established.** The current artifact does not establish a transition in a learned image model. **Next falsifier.** Freeze the oracle calibration, train independent learned scores, and evaluate speciation and nearest-training sample proximity on a held-out largest resolution.

9.2 Reinforcement learning

The state object is a policy/reward trajectory under KL pressure β_{KL} and verifier noise ϵ_{ver}. Gold reward, proxy reward, strategy support, KL, and entropy are recorded jointly because entropy reduction alone cannot distinguish productive specialization from reward collapse [13, 28].

Observation. The controlled map distinguishes a high-proxy/low-gold region from a supported-strategy region. **Interpretation.** The taxonomy expresses Goodhart behavior as joint robustness and commitment observables. **Not established.** It is not evidence for a phase transition in open RLVR training. **Next falsifier.** Repeat with independent environments, gold audits, process and outcome reward, and a prospective verifier-noise holdout.

9.3 Multi-agent populations

Intrinsic evidence h, social coupling J, topology, and population A are separate coordinates. Truth-conditioned consensus is the primary order; raw agreement is insufficient because a global false consensus can have maximal magnetization. Polarization, truth accuracy, message diversity, and error amplification are companion observables.

Observation. Controlled populations separate field-driven truth alignment, interaction-driven consensus, fragmentation, and amplified error. **Interpretation.** The PhaseCard prevents consensus from being equated with correctness. **Not established.** No shared Ising universality class or scaling law for deployed language-agent teams is claimed. **Next falsifier.** Vary population and topology prospectively with private evidence and a truth-labeled external task.

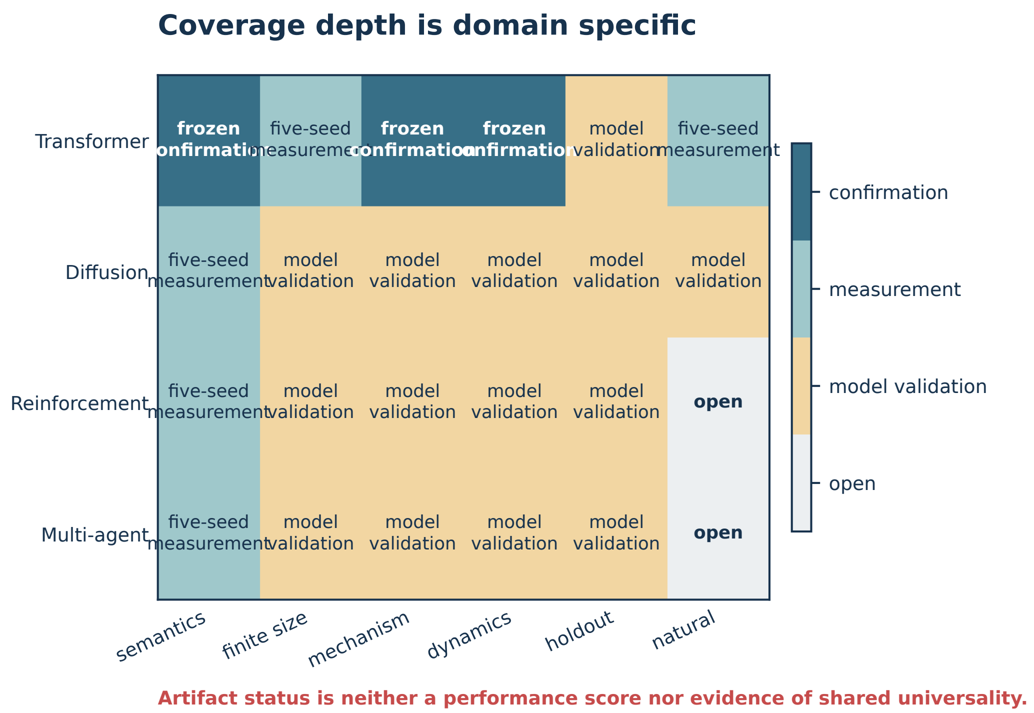

Figure 9: Artifact coverage and evidential depth. Transformer cells include five-seed measurements and frozen confirmations. Diffusion, reinforcement-learning, and multi-agent cells are mainly controlled model validation. Open natural/learned endpoints remain visible. Color is evidence depth, not performance and not a universality score.

10 Synthesis: answers to the research questions

RQ1—semantic measurement. The controlled retrieval order responds to width and effective data, and its risk decomposition is auditable. This supports the measurement on the registered synthetic task. It does not prove that the same order survives natural language.

RQ2—finite-size boundary. The answer is negative on the public reference grid. There is no interior susceptibility peak, no Binder crossing, no attention specialization, and no prospective largest-size prediction. The correct outcome is finite-width learning response, not phase transition.

RQ3—mechanism. Matched ablations support a bounded feed-forward contribution. The frozen suite also separates the registered MLP variants in task order. The result is task-conditional and does not establish an architectural phase split.

RQ4—optimizer and scale path. Frozen slices show substantial optimizer-conditioned risk differences, and uncapped data paths outperform the linear data path at the registered endpoint. Because compute, update geometry, and all hyperparameters are not fully matched, the result remains a contrast within the atlas rather than a global method ranking.

RQ5—predictive transport. The controlled continuation simulator verifies that calibration, censoring, baselines, and disjoint holdouts can be executed. Its generated response law makes near-exact prediction unsurprising; it is therefore not used as empirical evidence for transport in trained systems. A valid positive answer requires the dense Transformer experiment to predict an untouched size or realism edge.

RQ6—external scope. Natural corpora support tokenizer-independent bits-per-byte measurement on disjoint ranges. Diffusion, RL, and multi-agent artifacts demonstrate protocol portability. Learned diffusion endpoints, open RLVR, open-weight large models, and truth-labeled agent populations remain open. External validity is therefore partial and explicitly bounded.

10.1 What the taxonomy changes in practice

Three distinctions drive these answers. First, nominal control is separated from effective control, revealing the duplicated data point. Second, causal intervention evidence is separated from phase evidence, so a real MLP effect does not manufacture a Binder crossing. Third, artifact coverage is separated from evidential depth, so a cross-domain simulator does not become a natural endpoint merely because it shares plotting code.

The result is a coherent claim ledger rather than twenty independent figures. Each positive statement has a neighboring blocked statement and a next falsifier. This makes negative evidence cumulative: a future study can change the status by filling a named missing coordinate rather than restarting the terminology.

11 Discussion, limitations, and threats to validity

11.1 Scientific scope

The strongest trainable evidence concerns small decoder models on controlled retrieval and byte-level language modeling. It does not directly cover pretrained open-weight language models, natural images, open RLVR, or deployed agent populations. The cross-domain dossiers are model-validation cases and are not coequal replications. A seven-coordinate taxonomy improves the precision of these boundaries but cannot create missing external evidence.

The framework also does not assume that every system has an equilibrium Gibbs measure or a scalar thermodynamic limit. Optimizer trajectories, diffusion paths, feedback loops, and changing populations are non-equilibrium objects. The appropriate result may be a crossover, dynamical regime, path dependence, or censoring rather than a phase transition.

11.2 Reference-grid limitations

The GPU reference has only three widths, three nominal controls, five seeds, and 24 update steps. One nominal control pair collapses to the same effective data count, and all conditions reach the end of the short training horizon before the registered time-to-order criterion. These features limit both control resolution and dynamical interpretation. They are now shown directly rather than hidden by a smooth line.

Outer-seed intervals quantify initialization/data/minibatch variability under the registered generator but do not establish independent implementation replication. With only five seeds, susceptibility intervals are wide and power to detect small OOD penalties is limited. The negative phase result applies to the tested grid; it is not a proof of absence at larger width or longer time.

11.3 Exploratory multiplicity and confirmation

The broad tensor evaluates many families and observables. Its five-seed intervals are descriptive screens, not multiplicity-corrected hypothesis tests. The twelve-seed suite protects selected numerical contrasts with fresh seeds, but its hypotheses were chosen after inspecting the screen. It is a second-stage confirmation, not an independent external replication.

Optimizer contrasts use frozen learning-rate slices but do not exhaustively match update norm, noise scale, training compute, or tuning budget. Corpus bits per byte are tokenizer independent, yet corpora differ in intrinsic predictability and preprocessing. Neither panel should be read as a universal ranking.

11.4 Construct validity

Performance-derived order remains a task order unless a semantic probe connects it to a latent or functional property. Attention entropy is not strategy entropy; spectral effective rank is not macrostate multiplicity; and temporal checkpoint variance is not outer-disorder susceptibility. The observable dictionary reduces these errors, but any misspecified latent probe or null can still compromise construct validity.

Finite-size forms are empirical candidates outside solvable anchors. Width, depth, data, context, trajectory resolution, horizon, and population need not share an exponent. Data collapse can be overfit when too many exponents and windows are selected on the same sizes; the atlas therefore requires a prospective largest-size check before using phase language.

11.5 Broader use and non-goals

The atlas is intended to make claims easier to compare, reproduce, and falsify. It should not be used to lend physical prestige to a benchmark curve or to rank systems with a single evidence score. Theory, fidelity, evidence, and outcome remain vectors because collapsing them would erase the very distinctions the framework is designed to preserve.

12 Conclusion

We presented an evidence-aware statistical-mechanics atlas for numerical studies of modern machine learning. Seven scientific coordinates state what is being studied; a PhaseCard states how it is measured, perturbed, and held out; and an independent evidence record states what may be claimed. This separation turns semantic failure, censoring, unresolved contrasts, duplicated controls, and model misspecification into reportable outcomes rather than discarded runs.

The empirical program demonstrates why this discipline matters. Small Transformers show clear width-, data-, module-, optimizer-, and corpus-dependent responses, with frozen confirmations for selected contrasts. Yet the public GPU reference fails the finite-size gates required for a phase boundary. Diffusion, reinforcement-learning, and multi-agent dossiers show how trajectory, feedback, and population observables fit the same grammar without asserting shared universality. The immediate next step is prospective: execute the registered dense Transformer grid, hold out its largest width, and let prediction, crossing, or censoring determine the outcome. More broadly, the atlas provides a common route from suggestive plots to explicit ensembles, falsifiers, bounded claims, and cumulative negative knowledge.

Field	Operational definition	Invalid shorthand	Required record
System D	update and interaction law defining the domain	"neural network" without task/dynamics	domain and implementation family
State X	random object on which the observable is defined	mixing parameters, activations, and trajectories	shape, checkpoint/endpoint, transformation
Deformation δ	registered operator changing one scientific coordinate	"more realistic"	source, target, coordinates held fixed
Ensemble Ω	probability law and nesting of disorder	"five runs"	outer unit, inner replicas, seed streams
Scale $\mathbf{N}$	vector plus one declared finite-size coordinate and path	"model size"	values, path, caps, effective resources
Observable m	estimator, normalization, units, ensemble, null, range	"entropy" or "accuracy" alone	typed estimate and validity status
Phenomenon H	falsifiable collective-behavior hypothesis	visual phase label	diagnostics, alternative, next falsifier

Table 4: Formal dictionary for the scientific taxonomy. Protocol and evidence fields are recorded separately.

Observable family	Estimator examples	Necessary control	Common invalid substitute
Predictive risk	normalized CE/Brier; bits per byte; gold reward	chance, calibration, IID/OOD split	raw loss across incompatible tasks
Semantic recovery	latent probe, counterfactual accuracy, mutual information	shuffled latent, nuisance-matched probe	terminal performance alone
Specialization	component assignment, role occupancy, permutation-aligned overlap	permuted heads/roles, matched capacity	unaligned average attention
Multiplicity	correlation participation, overlap distribution, effective macrostate count	independent replicas and correlation null	spectral rank called a state count
Response	derivative, intervention response, peak width	window, resolution, and scale sensitivity	largest endpoint slope
Fluctuation	outer-seed susceptibility, Binder cumulant	independent terminal outer seeds	checkpoint variance in one run
Dynamics	time-to-order, commitment, path entropy, hysteresis	matched compute, forward/reverse path	final loss
Mechanism	raw and bounded matched-ablation risk effect	sham intervention, resource matching	ratio with vanishing denominator
Robustness	OOD risk, gold/proxy gap, truth consensus	held-out environment, truth labels	proxy score or agreement alone
Resource response	marginal quality per FLOP/token/NFE/message	equal-resource policy	wall time on unmatched hardware

Table 5: Observable catalog. Each row names a different estimand and a common failure mode.

A Taxonomy dictionary and edge semantics

An atlas edge applies one deformation while preserving an explicit source and target tuple. If multiple coordinates change, the edge is composite and the interaction

$$\Delta_{ij} g_{\mathrm{c}} = g_{\mathrm{c}}^{ij} - g_{\mathrm{c}}^{i} - g_{\mathrm{c}}^{j} + g_{\mathrm{c}}^{0} \tag{10}$$

can diagnose non-additivity. A composite edge is not evidence that either single-coordinate transport succeeded. Missing, unimplemented, attempted, invalidated, censored, and supported cells are distinct machine states.

B Observable and null catalog

Three "effective number" constructions illustrate the naming rule:

$$K_{\mathrm{eff}}^{\mathrm{macro}} = \exp H(Z), \qquad N_{\mathrm{eff}}^{\mathrm{corr}} = \frac{(\sum_i w_i)^2}{\sum_{ij} w_i w_j \rho_{ij}}, \qquad r_{\mathrm{eff}}^{\mathrm{spec}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \tag{11}$$

They count macrostates, correlation-adjusted units, and spectral participation, respectively. None can be substituted for another because the numerical range looks similar.

C Representative PhaseCards

Field	GPU reference	MLP confirmation	Natural endpoint	Multi-agent stress test
System/state	Transformer representation	Transformer representation	byte-model learning endpoint	beliefs/messages/actions
Deformation	effective data and width	feed-forward block	corpus	field, coupling, topology
Outer unit	trained model	trained model	trained model	population instance
Primary estimand	semantic order, IID risk	matched MLP risk effect	bits per byte	truth-conditioned consensus
Scale path	width; fixed depth/context	width and matched FF budget	data and model size	population and rounds
Critical null	shuffled/control task	sham and full-model evaluation	disjoint byte ranges	global false consensus
Holdout	none in pilot	fresh seeds and frozen variants	held-out ranges	controlled field/coupling cells
Outcome	finite-width response	bounded module contrast	measurement transport	model validation
Blocked claim	phase boundary	universal MLP ranking	corpus quality ranking	Ising universality

Table 6: Representative PhaseCards in human-readable form. Machine records also store nested seeds, hashes, resources, censoring, and invalidation.

D Registered dense Transformer confirmation

The next experiment is frozen in a portable configuration. It addresses the specific failures of the reference pilot rather than adding arbitrary compute:

- controls $\alpha \in \{2, 3, 4, 6, 8, 12\}$, which yield distinct effective sample counts at every width;
- widths $d \in \{16, 32, 64, 96, 128\}$, with $d = 128$ reserved for prospective evaluation;
- twelve fresh outer seeds per condition and no inner sample promoted to an outer replicate;
- 192 update steps with checkpoints every eight steps, so time-to-order and optimization censoring can be diagnosed;
- an example cap above the largest requested value, plus explicit requested/effective example fields;
- primary semantic order and IID risk; secondary OOD risk, MLP and attention intervention effects, attention entropy, susceptibility, Binder cumulant, gradient geometry, and trajectory observables; and
- a frozen decision: phase language requires an interior response feature, robust finite-size behavior below the holdout, and a successful $d = 128$ prediction. Otherwise the outcome is rounded, censored, or unresolved.

The design contains 360 trained models. It is not pooled with the pilot because training horizon and control support differ. Execution environment and scheduler settings are runtime inputs and are absent from the scientific configuration.

E Figure and manuscript contract

Every manuscript plot is generated at exactly 6.4×4.8 inches, saved as vector PDF and 300-dpi PNG, and uses one palette, marker order, font hierarchy, and panel-label convention. Exact physical size is preserved by avoiding tight-bounding-box resizing. Figure captions state the estimand, uncertainty, and blocked interpretation rather than repeating only axis labels.

The build places a float barrier at the end of every major section and clears all floats before appendices and references. This prevents figures from leaking into the bibliography or being separated from the claim they support. Tables use flexible-width columns and no manual scaling that would make text smaller than the caption.

F Artifact audit and reproducibility contract

F.1 Material findings

The audit distinguishes task count from scientific validity. The completed screen contains 8,700 trained models and the frozen follow-up 1,632, but those counts do not upgrade a performance-derived order, censored threshold, tuned endpoint, or overlapping split. Five material issues are recorded:

1. a legacy MLP causal ratio divided by a vanishing scale;
2. threshold endpoints were previously eligible to be counted as crossings;
3. optimizer learning rates were selected on an endpoint later reported;
4. one quadratic data path hit a training-example cap; and
5. two nominal reference controls collapse to the same effective 32-example condition at width 16.

The first four motivated the frozen confirmation suite; the fifth is exposed in Figure 7 and motivates the dense registered design.

F.2 Artifact identity and completeness

Scientific, protocol, implementation, and execution hashes are separate. Artifacts store condition identity, outer and nested seeds, raw outer values, interval method, metric units and validity, requested and effective resources, trajectory checkpoints, crossing/censor and cap-hit status, and data revision or content hash. Writes are atomic. Strict aggregation fails on missing seeds, non-finite required fields, incompatible trajectories, or mismatched schemas.

Legacy artifacts remain available but are not relabeled with statistics they did not record. If an estimand, split, ensemble, or selection rule changes, the old artifact is marked invalid for the affected claim. It is never silently repaired after inspection.

F.3 Portability and privacy

No public source, configuration, figure, PDF metadata, or evidence bundle stores a hostname, scheduler partition, user directory, private data location, or site-specific container path. Repository, manifest, output, data, interpreter, and accelerator settings are supplied at runtime. The public workflow can run locally or under a scheduler without changing the PhaseCard.

F.4 Release checklist

A release is valid only if all generated figures and result macros predate the compiled PDF; the bibliography resolves; the LaTeX log contains no overfull box; every PDF page is rendered and visually inspected; all plots have the registered physical size; source and artifacts pass path/hostname scans; and code tests, lint, and documentation checks pass. The committed screenshots are render evidence, not scientific data.

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